RELATION & FUNCTION

- **Relation**: Let A and B are two sets. A relation R from set A to set B is a subset of $A \times B$. So $R \subseteq A \times B$. It is denoted by $R: A \to B$ and $R = \{(a,b) : a \in A, b \in B\}$
  If $a \in A$ and $b \in B$ then:
  (i) If $(a,b) \in R$ means 'a is related to b' and is written as aRb.
  (ii) If $(a,b) \notin R$ means 'a is not related to b' and is written as $a\not{R}b$.

- **Domain, Co-domain and Range of a relation**: If R is a relation from set A to set B, then
  (i) Domain of R = $\{a : (a,b) \in R\}$.
  (ii) Range of R = $\{b : (a,b) \in R\}$.
  (iii) Co-domain of R = Set B.

- **Inverse of a Relation**: If $R: A \to B$, then its inverse is defined as a relation $R^{-1}: B \to A$, where $R^{-1} = \{(b,a) : (a,b) \in\}$.

- **Types of Relations**:

  (i) **Empty Relation or Void Relation**: $R: A \to A$, is an empty or void relation if $R = ¤$.
  (ii) **Universal Relation**: $R: A \to A$, is a universal relation if $R = A \times A$.
  (iii) **Identity Relation**: $R: A \to A$, is an identity relation if $R = \{(a,a) : a \in A\}$
  (iv) **Reflexive Relation**: $R: A \to A$, is reflexive if $(a,a) \in R$ for all a.
  (v) **Symmetric Relation**: $R: A \to A$, is symmetric if $(a,b) \in R$ then $(b,a) \in R$
  (vi) **Transitive Relation**: $R: A \to A$, is transitive if $(a,b) \in R$, $(b,c) \in R$ then $(a,c) \in R$.
  (vii) **Equivalence Relation**: $R: A \to A$, is reflexive if R is reflexive, symmetric and transitive.
  (viii) **Anti-Symmetric Relation**: $R: A \to A$, is anti-symmetric if $(a,b) \in R$ and $(b,a) \in R$ then $a = b$.
  (ix) **Composition of Relation**: $R: A \to B$ and $S: B \to C$, then composition of both the relation $R \circ S\{(a,c) : \exists b \in B, (a,b) \in R \text{ and } (b,c) \in S\}$.
  (x) **Congruence Modulo**: 'a Congruence to b modulo m' can be written as $a \equiv b (\bmod m)$ means $a - b$ is exactly divisible by m.

- **Number of relations**:

  (i) Let R is a relation from set A to set B. $|A| = m$ and $|B| = n$.
  Then number of relations = 2^{mn}.
  (ii) If $R: A \to A$ and $|A| = n$,
  (a) Then Number of relations = 2^{n^2}.
  (b) Then Number of reflexive relations = $2^{n^2 - n}$.

(c) Then Number of symmetric relations = $2^{\frac{n(n+1)}{2}}$.

(d) Then Number of anti-symmetric relations = $2^n 3^{\frac{n(n-1)}{2}}$.

(e) Then Number of both reflexive and symmetric relations = $2^{\frac{n(n-1)}{2}}$.

(f) Then Number of both reflexive and anti-symmetric relations = $3^{\frac{n(n-1)}{2}}$.

(g) Then Number of equivalence relations = Bell numbers B_n. Where

$B_n =$ 1, 2, 5, 15, 52, 203, 877, 4140, 21147, 115975,…

- **Function or Mapping**: Let X and Y be any two non-empty sets and there be correspondence and association between the elements of X and Y such that for every element of $x \in X$, there exists a unique element $y \in Y$, written as $y = f(x)$. Then we say that f is mapping or function from X to Y and is written as $f : X \to Y$ such that $y = f(x)$, $x \in X$ and $y \in Y$.
- **Real Function**: If $f : X \to Y$ be a function from a non empty set X to another non empty set Y, where $X, Y \subseteq R$, then we say that f is a real valued function or real function.
- **Features of a Mapping $f : X \to Y$**:

(1) For each element $x \in X$, there exist a unique element $y \in Y$.

(2) The element $y \in Y$ is called as the image of $x \in X$ under the mapping f.

(3) If there is an element in X which has more than one image in Y, then $f : X \to Y$ is not a function. But distinct element of X may be associated to the same element of Y.

(4) If there is an element in X which does not have an image in Y, then $f : X \to Y$ is not a function.

(5) The number of a functions that can be defined from X to Y is $(n(Y))^{n(X)}$.

- **Value of a function**: The value of a function $y = f(x)$ at $x = a$ is denoted by $f(a)$. It is obtaining by putting $x = a$ in $f(x)$.
- **Domain, Range and Co-domain of a Function**: If $f : X \to Y$ be a function, then the set X is said to be domain of f and range of f is set of all image points in Y under the mapping f.

  i.e. Domain of f = the set X.

  Range of f = the set $\{f(x) : f(x) \in Y, x \in X\}$

  Co-domain of f = the set X.
- **Different Types of Functions**:

(1) **One-one or Injective Function**: A function $f : X \to Y$ is said to be one-one or injective function if distinct elements of X have distinct image in Y.

(2) **Number of one-one functions**: If X and Y are any two finite sets with $|X| = m$ and $|Y| = n$, then number of one one function from X to Y = $\begin{cases} {}^nP_m & n \ge m \\ 0 & n < m \end{cases}$.

(3) **Many-one Function**: A function $f : X \to Y$ is said to be many-one function if there exists atleast two elements in X whose images are same.

(4) **Number of many-one functions**: If X and Y are any two finite sets with $|X| = m$ and $|Y| = n$, then number of many-one function from X to Y = $n^m - {}^nP_m$, $n \ge m$.

(5) **Rules to Check whether the function is one-one or many-one**:

(a) $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$, then $f(x)$ is one-one function.
(b) If a function is either strictly increasing or strictly decreasing in the whole domain (or equivalently, $f'(x) > 0$ or $f'(x) < 0$, $\forall x \in X$), then it is one-one, otherwise it is many-one.
(c) If any straight line parallel to x-axis intersects the graph of the function atmost at one point, then the function is one-one, otherwise it is many one. (i.e. it intersects the graph of the function in atleast two points).
(d) Any continuous function which has atleast one local maxima or local minima is many-one function.
(e) All even functions are many-one functions.
(f) All polynomials of even degree defined on *R* have atleast one local maxima or minima and hence are many-one functions on the domain *R*. Polynomials of odd degree can be one-one or many one.

(6) **Onto or Surjective function**: A function $f : X \to Y$ is said to be onto or surjective if every element of Y is the image of some elements of X under the map f.
NB: Any Polynomial of odd degree is an onto function.

(7) **Number of Onto functions**:

(a) If X and Y are any two finite sets with $|X| = m$ and $|Y| = n$, then number of onto function from X to Y
$= {}^nC_0 n^m - {}^nC_1(n-1)^m + {}^nC_2(n-2)^m - {}^nC_3(n-3)^m + {}^nC_4(n-4)^m - \ldots$
(b) If X and Y are any two finite sets with $|X| = m \ge 2$ and $|Y| = 2$, then number of onto function from X to Y = $2^m - 2$.

(8) **Into Function**: A function $f : X \to Y$ in an into function if it not onto function.
NB: Any Polynomial of even degree is an into function.

(9) **Number of Into functions**: If X and Y are any two finite sets with $|X| = m$ and $|Y| = n$, then number of into function from X to Y
$= {}^nC_1(n-1)^m - {}^nC_2(n-2)^m + {}^nC_3(n-3)^m - {}^nC_4(n-4)^m + \ldots$

(10) **Rules to Check whether the function is Onto or Into**:

(a) Let a function $f : X \to Y$ is given. Find the range of the function. If range of the function is Y, then function is onto otherwise it is into function.

(11) **Bijective function**: A function $f : X \to Y$ is said to be bijective, if it is both one-one and onto.

(12) **Number of Bijective functions**: If X and Y are any two finite sets with $|X| = n$ and $|Y| = n$, then number of bijective functions from X to Y = $n!$.

(13) **Constant Functions**: A function $f : R \to R$ defined as $f(x) = c, \forall x \in R$, where c is a constant, is called as constant function. Its domain is R and range is {c}. The graph of constant function is a straight line parallel to x-axis when x is independent variable.

(14) **Identity function**: The function $f : R \to R$ defined as $f(x) = x, \forall x \in R$, is called as identity function. Its domain is R and range is R.

(15) **Equality of functions**: Two functions f and g are said to be equal if:

(a) Domain of f = Domain of g.
(b) Range of f = Range of g.
(c) $f(x) = g(x), \ \forall x \in domain$.

(16) **Modulus function or Absolute value function**: The function $f : R \to R$, is defined as $f(x) = |x| = \begin{cases} x & x > 0 \\ 0 & x = 0 \\ -x & x < 0 \end{cases}$ is called as modulus function or absolute value function.

Its domain is R and range is $R^+ \cup \{0\}$ or $[0, \infty)$.

(17) **Properties of Modulus function**:

(a) $|x + y| \le |x| + |y|$.
(b) $|x + y| \ge \big||x| - |y|\big|$.

(18) **Greatest Integer function or Step function or Integral Function**: The function $f : R \to R$, is defined as $f(x) = [x]$ is called as greatest integer function where [x] is the integral part of x or greatest integer not greater than x or greatest integer less than or equal to x. or
[x] = x if x is an integer.
[x] = left side nearest integer if x is not an integer.

(19) **<u>Properties of Greatest Integer function</u>**:

(a) $[x] \le x \le [x]+1$.

(b) $[x+y] = \begin{cases} [x]+[y] & if\ \{x\}+\{y\}<1 \\ [x]+[y]+1 & if\ \{x\}+\{y\} \ge 1 \end{cases}$ where {x} means fractional part of x.

(c) $n \le x < n+1 \Leftrightarrow [x] = n$.

(d) $x-1 < [x] \le x$.

(e) $[[x]] = [x]$.

(f) $[n+x] = n+[x]$ where n is an integer.

(g) $[x]+[-x] = \begin{cases} 0 & if\ x \in Z \\ -1 & if\ x \notin Z \end{cases}$.

(20) **<u>Fractional Part Function</u>**: The function $f: R \to R$, is defined as $f(x) = \{x\} = x-[x]$, is called as fractional part function. Its domain is R and range is [0, 1). (here {x} denotes fractional part of x)

(21) **<u>Properties of Fractional part function</u>**:

(a) If x is an integer then $\{x\} = 0$.

(b) $[\{x\}] = 0$.

(c) $0 \le \{x\} < 1$.

(d) $\{x\}+\{-x\} = \begin{cases} 0 & if\ x \in Z \\ 1 & if\ x \notin Z \end{cases}$

(22) **<u>Signum Function</u>**: The function $f: R \to R$, is defined as $f(x) = \begin{cases} \frac{|x|}{x} & x \ne 0 \\ 0 & 0 \end{cases}$, or

$$f(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$ is called as signum function.

Its domain is R and range is {-1, 0, 1}.

(23) **<u>Reciprocal function</u>**: The function $f: R-\{0\} \to R$, is defined as $f(x) = \frac{1}{x}$, is called as reciprocal function. Its domain is $R-\{0\}$ and range is $R-\{0\}$.

(24) **<u>Exponential function</u>**: Let $a \ne 1$ be a positive real number. Then the function $f: R \to R$, defined as $f(x) = a^x$ is called as exponential function. Its domain is R and its range is $(0, \infty)$.

(25) **Logarithmic function**: Let $a \neq 1$ be a positive real number. Then the function $f:(0,\infty) \to R$, defined as $\log_a x$ is called as logarithmic function. Its domain is $(0,\infty)$ and range is R.

(26) **Polynomial function**: A function $f:R \to R$ defined by $f(x)=a_0+a_1x+a_2x^2+...+a_nx^n$, where $n \in N$ and $a_0,a_1,a_2,...,a_n \in R$ is called as polynomial function. If $a_n \neq 0$, then it is called as polynomial of degree n. Its domain is R.

(27) **Rational function**: A function of the form $f(x)=\dfrac{A(x)}{B(x)}$, where $A(x), B(x)$ are polynomials over the set of real numbers and $B(x) \neq 0$, is called as rational function. Its domain is $R-\{x:B(x)=0\}$.

(28) **Finding Domains of the function**: Let f and g are two real functions defined with domain D_1 and D_2. Then:

(a) The sum function $(f+g)$ is defined by $(f+g)(x)=f(x)+g(x)$, $\forall x \in D_1 \cap D_2$. The domain of $(f+g)$ is $D_1 \cap D_2$.

(b) The difference function $(f-g)$ is defined by $(f-g)(x)=f(x)-g(x)$, $\forall x \in D_1 \cap D_2$. The domain of $(f-g)$ is $D_1 \cap D_2$.

(c) The product function (fg) is defined by $(fg)(x)=f(x)g(x)$, $\forall x \in D_1 \cap D_2$. The domain of (fg) is $D_1 \cap D_2$.

(d) The quotient function $\dfrac{f}{g}$ is defined by $\left(\dfrac{f}{g}\right)(x)=\dfrac{f(x)}{g(x)}$, $\forall x \in D_1 \cap D_2-\{x:g(x)=0\}$. The domain of $\dfrac{f}{g}$ is $D_1 \cap D_2-\{x:g(x)=0\}$.

(e) The scalar multiple function cf is defined by $(cf)(x)=cf(x)$, $\forall x \in D_1$. The domain of cf is D_1.

(29) **Composition of functions**: Let f and g are two real functions defined with domain D_1 and D_2.

(a) If range of $f \subseteq$domain of g, then the composite function $(g \circ f)(x)=g(f(x))$, $\forall x \in D_1$.

(b) If range of $g \subseteq f$ domain of, then the composite function $(f \circ g)(x)=f(g(x))$, $\forall x \in D_2$.

(30) **Useful results on composition of functions**: If $f:X \to Y$ and $f:Y \to Z$ then

(a) If both f and g are one-one then $(g \circ f)$ is also one-one.
(b) If both f and g are onto then $(g \circ f)$ is also onto.
(c) If $(g \circ f)$ is one-one, then f is one-one but g may not be one-one.
(d) If $(g \circ f)$ is onto, then g is onto but f may not be onto.
(e) If both f and g are bijective then $(g \circ f)$ is also bijective.
(f) It may happens that $(g \circ f)$ may exist and $(f \circ g)$ may not exist. Moreover, even if both $(g \circ f)$ and $(f \circ g)$ exists, they may not be equal.

(31) **Inverse function**: If the function $f : X \to Y$ is both one-one and onto i.e. bijective then we define inverse function $f^{-1} : Y \to X$ by the rule $y = f(x) \Rightarrow x = f^{-1}(y),\ \forall x \in X, \forall y \in Y$.

(32) **Rules to find the inverse of function**: Let $f : X \to Y$ is a bijective function.

(a) Put $f(x) = y$.
(b) Solve the equation $y = f(x)$ to obtain x in terms of y. Interchange x and y to obtain the inverse of f.
(c) The graph of $y = f(x)$ and its inverse written in the form $x = g(y)$ are symmetrical about the line $y = x$.
(d) The graph of $y = f(x)$ and $y = f^{-1}(x)$, if intersects then meet on the line $y = x$ only.
(e) The solution of $f(x) = f^{-1}(x)$ is also solution of $f(x) = x$.
(f) If $f : X \to Y$ and $g : Y \to Z$ are two bijective functions, then $g \circ f : X \to Z$ is bijective and $(g \circ f)^{-1} = (f^{-1} \circ g^{-1})$.

(33) **Even extensions**: A function $f(x)$ defined on the interval $[0,a]$ can be extended $[-a,a]$, so that $f(x)$ becomes an even function on the interval $[-a,a]$, then this extension is called as even extension.

Let $x(x)$ be the even extension of $f(x)$, then $g(x) = \begin{bmatrix} f(-x) & x \in [-a,0] \\ f(x) & x \in [0,a] \end{bmatrix}$.

(34) **Odd extensions**: A function $f(x)$ defined on the interval $[0,a]$ can be extended $[-a,a]$, so that $f(x)$ becomes an odd function on the interval $[-a,a]$, then this extension is called as odd extension.

Let $x(x)$ be the odd extension of $f(x)$, then $g(x) = \begin{bmatrix} -f(-x) & x \in [-a,0] \\ f(x) & x \in [0,a] \end{bmatrix}$.

(35) **Even function**: A function $f(x)$ is said to be an even function if $f(-x) = f(x)$ for every real number x in the domain of $f(x)$.

(36) **Odd function**: A function $f(x)$ is said to be an odd function if $f(-x) = -f(x)$ for every real number x in the domain of $f(x)$.

(37) **Properties of Odd and Even function**:

(a) The graph of odd function is symmetric about origin i.e. symmetric about opposite quadrants.
(b) The graph of even function is symmetric about the y-axis.
(c) A function $f(x)+f(-x)$ always represents an even function.
(d) A function $f(x)-f(-x)$ always represents an odd function.
(e) Any function $f(x)$ can be represented as sum of one even and one odd function. $f(x)=\frac{f(x)+f(-x)}{2}+\frac{f(x)-f(-x)}{2}$.
(f) $f(x)=0$ is the only function which is even and odd both.
(g) If $f(x)$ is an odd function and $f(x)$ is differentiable then $f'(x)$ is an even function.
(h) If $f(x)$ is an even function and $f(x)$ is differentiable then $f'(x)$ is an odd function.
(i) If f and g are even functions then $f \circ g$ is also even provided $f \circ g$ is defined and exist.
(j) If f and g are odd functions then $f \circ g$ is also odd provided $f \circ g$ is defined and exist.
(k) If f is an even function and g is odd function then $f \circ g$ is even provided $f \circ g$ is defined and exist.
(l) If f is an odd function and g is even function then $f \circ g$ is even provided $f \circ g$ is defined and exist.

(38) **Periodic function**: A function $f(x)$ is said to be a periodic function, if there exist a real number $T > 0$, such that $f(x+T) = f(x)\ \forall x \in R$. The smallest real number $T > 0$, satisfying $f(x+T) = f(x)\ \forall x \in R$ is called as fundamental period of $f(x)$.

(39) **Rules to find periodicity of function**:

(a) Put $f(x) = f(x+T)$ and solve this equation to find the positive value of T independent of x.
(b) If no positive values of T independent of x is obtained then $f(x)$ is non periodic function.
(c) If positive values of T independent of x are obtained then $f(x)$ is periodic function and the least positive value of T is the fundamental period of the function $f(x)$.
(d) Constant function is periodic with no fundamental period.

(e) If $f(x)$ is T periodic, then $\frac{1}{f(x)}$ is also T periodic.

(f) If $f(x)$ is T periodic, then $\sqrt{f(x)}$ is also T periodic.

(g) If $f(x)$ is T_1 periodic and $g(x)$ is T_2 periodic, then $f(x)+g(x)$ is of period equal to $lcm(T_1,T_2)$, provided there is no positive k such that $f(x+k)=g(x)$ and $g(x+k)=f(x)$.

(h) If $f(x)$ is T periodic, then $kf(ax+b)$ is $\frac{T}{|a|}$ periodic, where $a,b,k \in R$ and $a,k \neq 0$.

(i) $f(x)=\sin x$ is of period 2® .

(j) $f(x)=\cos x$ is of period 2® .

(k) $f(x)=\tan x$ is of period ® .

(l) $f(x)=\cot x$ is of period ® .

(m) $f(x)=\sec x$ is of period 2® .

(n) $f(x)=\cos ecx$ is of period 2® .

(o) $f(x)=|\sin x|$ is of period ® .

(p) $f(x)=|\cos x|$ is of period ® .

(q) $f(x)=|\tan x|$ is of period ® .

(r) $f(x)=|\cot x|$ is of period ® .

(s) $f(x)=|\sec x|$ is of period ® .

(t) $f(x)=|\cos ecx|$ is of period ® .

(u) $f(x)=\sin^n x$ is of period 2® if n is odd and $f(x)=\sin^n x$ is of period ® if n is even.

(v) $f(x)=\cos^n x$ is of period 2® if n is odd and $f(x)=\cos^n x$ is of period ® if n is even.

(w) $f(x)=\tan^n x$ is of period ® .

(x) $f(x)=\cot^n x$ is of period ® .

(y) $f(x)=\sec^n x$ is of period 2® if n is odd and $f(x)=\sec^n x$ is of period ® if n is even.

(z) $f(x)=\cos ec^n x$ is of period 2® if n is odd and $f(x)=\cos ec^n x$ is of period ® if n is even.

(aa) If $f(x)$ is T periodic and $g(x)$ is any function such that domain of $f \subset$ domain of g, then $g \circ f$ is also periodic with period T.

(40) **<u>Rules to find Domain of the function</u>**:

(a) For Algebraic expression denominator should be non zero.

(b) For Algebraic expression under the even root should be non negative.

(c) $\log_a b$ is defined when $a > 0, b > 0$ and $b \neq 1$.

(d) a^x is defined for all real values of x, where $a > 0$.

(e) $(x-a)(x-b) > 0 \Rightarrow x < a \text{ or } x > b$ for $a < b$.

(f) $(x-a)(x-b) < 0 \Rightarrow a < x < b$ for $a < b$.

(g) $|x| < a \Rightarrow -a < x < a$.

(h) $|x| > a \Rightarrow x < -a \text{ or } x > a$.

(i) $\sqrt{x^2} = |x|$.

(j) $\sqrt[n]{x^n} = \begin{cases} |x| & n \text{ even} \\ x & n \text{ odd} \end{cases}$.

(k) $\log_a b > k \Rightarrow \begin{cases} b > a^k & \text{if } a > 1 \\ b < a^k & \text{if } a < 1 \end{cases}$.

(41) **Rules to find Range of the function**:

(a) Find the domain of the function $y = f(x)$.

(b) If the domain is an infinite interval, solve the equation $y = f(x)$ and find x in terms of y to get $x = g(y)$, find the real values of y for which x is real. The set of values of y so obtained is the range of $y = f(x)$. Note that if finite number of values of x are excluded from the domain, find the values of y for these values of x and exclude these values from the range of $y = f(x)$ found.

(c) If the domain is finite interval, find the least and greatest value of y for values of x in the domain. If a is the least value and b is the greatest value then range of $y = f(x)$ is [a, b].

(42) **General Results on function**: If x and y are independent variables, then:

(a) $f(xy) = f(x) + f(y) \Rightarrow f(x) = k\log_e x$ or $f(x) = 0$.

(b) $f(xy) = f(x) \times f(y) \Rightarrow f(x) = x^n$.

(c) $f(x+y) = f(x) \times f(y) \Rightarrow f(x) = a^{kx}$.

(d) $f(x+y) = f(x) + f(y) \Rightarrow f(x) = kx$.

(e) $f(x+y) = f(x) = f(y) \Rightarrow f(x) = k$, where k is a constant.

(f) A continuous function $f(x)$ takes only rational values for all $x \Rightarrow f(x)$ is a constant function.

(g) By considering a general nth degree polynomial and writing the expression $f(x) + f\left(\frac{1}{x}\right) = f(x) \times f\left(\frac{1}{x}\right) \Rightarrow f(x) = 1 \pm x^n$.

- **Intervals**: A subset of the real number set R is called as an interval.

(1) **Closed Interval**: $[a, b] = \{x \in R : a \le x \le b\}$.

(2) **Open Interval**: $(a,b)=\{x\in R: a<x<b\}$.
(3) **Left Open Right Closed Interval**: $(a,b]=\{x\in R: a<x\le b\}$.
(4) **Right Open Left Closed Interval**: $[a,b)=\{x\in R: a\le x<b\}$.
(5) $[a,\infty)=\{x\in R: x\ge a\}$.
(6) $(a,\infty)=\{x\in R: x>a\}$.
(7) $(-\infty,a]=\{x\in R: x\le a\}$.
(8) $(-\infty,a)=\{x\in R: x<a\}$.

- **Neighborhood of a point**: Let $a\in R$. If $¢>0$, then the open interval $(a-¢,\ a+¢)$ is called as $¢$ neighborhood of the point a. It is denoted by $N_{¢}(a)$. Here 'a' is called as centre and $¢$ is called as radius of the neighborhood $N_{¢}(a)$.
  $N_{¢}(a)=(a-¢,\ a+¢)=\{x\in R: a-¢<x<a+¢\}=\{x\in R: |x-a|<¢\}$
  The set $N_{¢}(a)-\{a\}$ is called as deleted neighborhood of the point 'a'.
  $(a-¢,\ a)$ is called as left $¢$ neighborhood of the point 'a'.
  $(a,\ a+¢)$ is called as right $¢$ neighborhood of the point 'a'.

- **Limit Point of a set**: Let A be a subset of R. A point 'a' is said to be a limit point of A if every nbd of 'a' contains at least one point of A other than 'a'.
  If 'a' is a limit point of a set A, it follows that every nbd of 'a' contains infinitely many points of A.

- **Monotonic Function**: Let $f: X\to Y$ be a function. Then f is said to be:

(1) **Monotonically Increasing** on X if $x_1,x_2\in X$, $x_1<x_2 \Rightarrow f(x_1)\le f(x_2)$.
(2) **Monotonically Decreasing** on X if $x_1,x_2\in X$, $x_1<x_2 \Rightarrow f(x_1)\ge f(x_2)$.
(3) **Strictly Increasing** on X if $x_1,x_2\in X$, $x_1<x_2 \Rightarrow f(x_1)< f(x_2)$.
(4) **Strictly Decreasing** on X if $x_1,x_2\in X$, $x_1<x_2 \Rightarrow f(x_1)> f(x_2)$.
(5) A **monotonic function** on X if it is either monotonically increasing or monotonically decreasing function on X.

- **Bounded Function**: A function $f: A\to R$ is said to be bounded function:
  If there exist numbers k_1,k_2 such that $k_1\le f(x)\le k_2,\ \forall x\in A$

  OR

  If there exist numbers k such that $|f(x)|\le k,\ \forall x\in A$
  Otherwise f is said to be unbounded.

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